

Name Student: Winnie Dao

Internship or Thesis: Internship

Title project: Augmented Reality: A Tool in (Service Mathematics) Education

Abstract (100 - 150 words):

A decrease in student motivation has been observed among students that have been subjected to the sudden shift to remote learning due to the COVID-19 pandemic. As student motivation is closely related to academic results, it is important that this problem is addressed. One way that has been proposed to increase motivation is through augmented reality (AR). AR has been shown by previous studies to increase both motivation and academic performance. A pilot study has been performed to investigate the students' attitude towards an AR app developed by The PRogramme of Innovation in Mathematics Education (PRIME) to support the service mathematics course that they are following. The pilot's result indicates that the students have a positive attitude towards the app: the students were satisfied with the application, more engaged in learning and feel that the app is an extension to and serves as enhancement of learning.

Name company or University: TU Delft

(If applicable) External supervisor: Jacqueline Wong

UU Supervisor: Leendert van Maanen

Name Student: Anahid Karimi

Internship or Thesis: Thesis (external)

Title project: Creatively creating intellectual property: Can personality tests predict whether someone is an innovator?

Abstract (100 - 150 words):

This study aims explore the relation between creativity and personality in order to provide ways to identify creative people within Signify who could potentially become part of their Innovation Team. More specifically we investigate relations between personality type and creativity, ideational behaviour, divergent and convergent thinking. Participants answered a total of 285 questions across three online questionnaires (The Big Five, the Runco Ideational Behavior Scale and the Remotes Associates Test) through Qualtrics. Participants comprised of two groups with either a psychology background (51 students) or a beta background (8 students). We expect to find a relation between Big Five factor 'Openness' and creative ability with prediction scores between 10 to 50 % based on available studies. Data analysis is currently still in progress.

Keywords: *Creativity, personality, ideational behaviour, divergent thinking, convergent thinking, RAT, RIBS, NEO-PI-R, Signify.*

Name company or University: Signify Netherlands

(If applicable) External supervisor: Tobias Borra

UU Supervisor: Caroline Junge

Name Student: Eleni Evangelatou

Internship or Thesis: Internship & Thesis (combined)

Title project: National Brands (A-brands) vs. Supermarket Brands (B-brands): Detecting bias in binary preference-based food decisions in the Netherlands”

The influence of branding on consumers’ choice behavior has been substantial. However, the psychological mechanisms underlying this influence have yet to be discovered. Initially inspired by the previous work of Philastides, M. G., & Ratcliff, R. (2013), we aim to investigate whether a bias is introduced in consumers’ choice behavior when making a binary choice between labeled food items in a typical Dutch supermarket, and measure it. My work so far has focused on setting up the experimental design and piloting; under time pressure, a two-alternative forced choice test (2AFC) is conducted that pairs two main brand types of food products, namely National Brands (or A-brands) and Supermarket Brands (or B-brands). A-brands are famous and established products manufactured and developed by a company and are distributed nationwide or internationally (e.g., Coca-Cola), while B-brands are owned and labeled by retailers and are exclusively available in their stores (e.g., Albert Heijn Cola). During the next period of my thesis, we will use computational modeling (Drift Diffusion Model) to analyze our data and either confirm or reject our hypothesis that branding itself affects choice behavior. Data are being currently gathered through Sona Systems.

Name company or University: Utrecht University

UU Supervisor: Dr. Martijn Mulder

Yasaman Javadi

Internship

Title: A review on possible gender differences in motion sickness

In this literature review, I am searching for possible factors that may be influencing higher susceptibility to motion sickness among women. The idea started in the number of literature where there was evidence that females are more prone to become motion sick while using technologies equipped with virtual reality. Among all the literature reviewed up to now, there is no definitive answer whether women are more susceptible to motion sickness or not. But considering today's progressive technology, the use of virtual reality is going to be much more common in near future. Therefore, it is important to investigate the presence of this possibility more in-depth in order to be sure whether 50% of the world's population truly is more sensitive to becoming motion sick while being in a special situation or not.

Utrecht University

UU Supervisor: Stella Donker

Pupillary response to brightness with peripheral brightness control

Pseudoneglect is defined as an attentional bias towards the left side of the visual field. Previous measures of pseudoneglect require an overt response and are sensitive to errors (Jewell & McCourt, 2000). The current paradigm uses pupillometry in combination with brightness variance to investigate this attentional bias. The pupil is known to constrict for bright stimuli and dilate for dark stimuli. Differences in pupil dilation between hemifields can show attentional biases (Strauch et al., in press). Black and white bars (on a grey background) were presented in various configurations while the participant attended the center. Bars could either be thin or wide and were presented centrally or peripherally. As hypothesized, centrally presented bars did not elicit a difference in pupil light response between black- and white configurations. Peripherally presented bars elicited a rightward bias, whereas a leftward bias was expected. The width of the bars showed no difference between conditions. The found rightward bias could be explained by the measured right eye, as an earlier effect of measured eye, though small, was found (Strauch et al., in press). Additional binocular testing will be done to further investigate the effect of the measured eye.

Name Student: Sara Lutgerink

Internship or Thesis: Thesis

Title project: Individual differences in emotion regulation – EEG study

Fear extinction is a process in which a fearful reaction to a specific trigger is unlearned by confrontation with that trigger without the feared consequence. The current study examined how theta and gamma oscillations of the prefrontal region relate to poor and normal fear extinction. The current study also examined age differences in neurological patterns of fear extinction. A group of 30 adolescents and young adults participated in a fear conditioning and recall task and an emotional regulation task. Scalp surface EEG was measured during the fear recall and emotional regulation tasks. Participants with poor fear extinction and participants with normal fear extinction are expected to show different levels of theta and gamma activity in the AMC and PFC. The current study examines the associations between age, emotion regulation activities and level of fear extinction.

Name company or University: Utrecht University

UU Supervisor: Joke Baas

Name Student: Sander van Wincoop

Internship or Thesis: Thesis

Title project: Top-down Influence on Skin Conductance Response

In a study by Binda et al. it was found that pupil size is influenced by a higher level visual analysis, taking content into consideration as well. This was demonstrated by seeing a decrease in pupil size when being shown a picture of the sun, even if the picture bears a decrease in overall light-level.

In my thesis, I am conducting an experiment where I will investigate if a similar response can be seen in skin conductance. Usually, skin conductance rises as temperature perception goes up. I am investigating if this also follows for simply seeing images that are associated with high temperatures. Should this indeed be the case, we would see a higher skin conductance response for images that are associated with high temperatures as opposed to images that are associated with cold temperatures or control images.

Name company or University: Utrecht University

(If applicable) External supervisor: n/a

UU Supervisor: Christoph Strauch

Name Student: Floris Geerts

Internship or Thesis: Internship

Title project: De invloed van automatisering van voertuigen op controlekamers

Om capaciteitsverhoging op het spoor te ondersteunen, gaat er geautomatiseerd worden. We hebben een literatuuronderzoek gedaan naar de invloed van automatisering van voertuigen op controlekamers. Een lage werkdruk kan zorgen voor een lagere situation awareness. Om dit te onderzoeken hebben we niet alleen gekeken naar literatuur binnen de treinsector, maar ook naar andere sectoren: de metro en luchtvaart. Vondsten vanuit deze sectoren hebben we vertaald naar de treinsector. We hebben gevonden dat de capaciteitsverhoging de werkdruk voor de treindienstleider verhoogt. De gepaarde automatisering om deze verhoging te ondersteunen verlaagd juist de werkdruk, omdat treinen voorspelbaarder rijden en door een vermindering in menselijke fouten. Zowel in de luchtvaart als bij de metro is het systeem in de controlekamer meer geïntegreerd. Dit zou ook aan te raden kunnen zijn voor de treinsector, wanneer er meer wordt geautomatiseerd.

Name company or University: ProRail

External supervisor: Alexander Admiraal

UU Supervisor: Sjoerd Stuit

Name Student: Andy Awa

Thesis

Title project: gaining insight into the opinion of Dutch donors concerning a number of relevant topics.

Abstract:

For my internship project I performed research at the market research company direct research in collaboration with the non-profit called "stichting donateursbelangen". This is a Dutch non-profit that fights for the right of charity donors. I performed an industry standard research following the process that is usual in this field of work. This consisted of me collaborating on a questionnaire, analyzing these results and reporting them to the client. The aim of this research was to gain more insight into the opinion of donors in the Netherlands with regards to donor privacy, transparency of charities and the implementation of a central app which allows donors to express their preference when it comes to charity contact. These insights could help to improve the donor centric fundraising in the Netherlands.

Name company: Direct research

External supervisor: Carlijn van de Pas

UU Supervisor: Roy Hessels

Name Student: Tom van Kempen

Internship or Thesis: Internship

Title project: Virtual reality as a novel child-friendly detector of visual field defects.

Abstract:

By causing scotomas in the visual field, cortical visual impairment affects multiple cognitive functions that rely on vision. The exact locations of these scotomas should be precisely detected to enable successful rehabilitation. However, standard automated perimetry (SAP), requires a certain level of concentration, fixation, task-comprehension and head stability, requirements that are unable to meet for young children. The current study assesses gaze-contingent flicker pupil perimetry (gcFPP) inside a virtual reality (VR) apparatus as an objective and passive form of VF testing and compared it to the method presented on a screen.

The results of the gcFPP on screen for healthy controls showed an increased pupil response when stimuli were presented in the temporal regions compared to the nasal regions. This nasal-temporal asymmetry was not observed in gcFPP inside a VR apparatus. None of the two methods detected the expected asymmetry between the superior and inferior regions. However, the overall pupil responses significantly correlated between the two methods, which shows opportunities.

Name company or University: University of Utrecht

UU Supervisor: Marnix Naber

Name Student: Margot van Muiswinkel

Internship or Thesis: Thesis

Title project: Visual communication systems: The use of gestures and digital communication systems in people with dementia.

Abstract (100 - 150 words):

One out of five people develop dementia. As a result, people often experience communication problems. Communication contributes to the quality of life as it enables people to express needs, wishes and feelings. The aim of this research is to investigate whether the use of gestures and digital communication systems can be used for people with dementia. This was investigated through literature review, interviews and fieldwork. The participants were acquired through a snowball effect. This research showed that the use of gestures is a very good communication technique. Same as digital communication tools, but then you miss human contact. This research showed that no general answer can be given to the research question, because of the fact that every person is unique and has different background and preferences. In the future, it is important to involve people with dementia in designing new communication tools by listening and looking at them.

Name company or University: University of Utrecht

UU Supervisor: Jessica Heeman

Name Student: Sebas van Hal

Internship or Thesis: Thesis

Title project: The extent of automaticity of gaze behavior during crowd navigation

Abstract (100 - 150 words):

The present research looks into the patterns of gaze behavior of participants who have been instructed to walk a route at a festival in Utrecht. Half of the participants have been instructed to avoid eye contact while navigating around the festival, the other half received no additional instruction, all participants wore eye-trackers while navigating around the festival.. Almost all research that has been conducted on people's gaze behavior has been performed using videos or photographs, and the current paradigm has been formed based on that. There is however no reason to assume that these findings translate well to the real life scenario. Results show that people do have deliberate control on their gaze behavior during crowd navigation, seeing as there was a significant difference in the amount of times people fixated on other's faces between the group that was instructed to avoid eye contact and the group that received no additional instruction.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Name Student: Sabien van der Sluis

Internship or Thesis: internship

Title project: do negative emotions work in advertisements? A literature review

In my internship, I look at the effects of using negative emotions in advertisements. My focus mainly lies on television commercials and charities. In my literature review I found that negative emotions can be a powerful tool, but it is not without its flaws. Negative emotions in advertisements can lead to psychological numbing to name just one side effect. Additionally, I also wrote articles for their website and supported the sales department by helping with the organisation of an event and proofreading other articles. I also reached out to potential new customers based on tests done on their commercials, and researched distinctive brand assets by creating a survey with brand logo's.

Name company or University: Validators

(If applicable) External supervisor: Jelmer Wit

UU Supervisor: Sjoerd Stuit

Name Student: Sara Pitkänen

Internship or Thesis: Thesis

Title project: The Cognitive Mechanisms of Nudging in the Context of Healthier Food Choices

Nudging refers to ways to influence one's decisions, however, the cognitive mechanisms of nudging are not fully understood. Suggested mechanisms are related to automatic and fast processing, where time pressure plays an important role. In this study, participants rated preferences for food items and made decisions between two food items under different conditions. We wanted to answer the following questions: will the participants choose the nudged item more often; does the time pressure strengthen the effect of the nudge; and will the ratings of the items influence the decisions? We first hypothesized that participants would select the nudged items more often, and that time pressure would increase this effect. Secondly, we expected that the preferences would influence these decisions. This study demonstrated that participants selected the nudged items more often, and there was a significant effect on the ratings of the items; meaning that people selected what they liked. Unexpectedly, time pressure did not influence these decisions. These results could have implications on how and where to use nudges effectively.

Name company or University: Utrecht University

UU Supervisor: Leendert van Maanen

Terrence te Loo – 5727154
Tanja Nijboer, Charlotte Southcombe
Jeroen Benjamins
Utrecht University – Internship

Mixed Reality in Cognitive Rehabilitation, What are the Possibilities?

The internship will start by conducting a survey for better appliance of MR in cognitive rehabilitation. Together with Laura Cibulskyte, an ACP master student, we constructed an online survey that is aimed to explore healthcare professionals and their attitudes towards the use of MR in cognitive rehabilitation. The data from this survey will be both qualitative and quantitative, and will not only explore thoughts and opinions on the usage of MR in cognitive rehabilitation, but it will also provide insight into expected barriers to the implementation.

The main objective of the internship will be to investigate the different views that healthcare professionals have towards the use of Mixed Reality in cognitive rehabilitation (i.e. cognitive assessment and training). The survey questions are designed to explore healthcare professionals and their attitudes towards the use of MR in cognitive rehabilitation, whilst other questions will focus more towards specific use cases that these technologies can be applied to.

Research questions: What are the potential barriers of using MR in cognitive rehabilitation?
How can MR be implemented correctly in the cognitive rehabilitation process?

Name Student: Gianfranco Bocanegra

Internship or Thesis: Thesis

Title project: Fear of Climate Change and its effect on Food Choice in young adults

In public health, fear appeals have been used, for example, to incentivize sunscreen products, for anti-smoking campaigns, to advertise a sleeping pill, vaccination, drug prevention, and to promote the use of preservatives evoking the threat of HIV infection.

Even though there are plenty of studies examining the impact of fear appeals from the health and marketing disciplines, fear appeal and their effect in changing diets, evoking environmental reasons on young adults are limited.

The purpose of the study is to determine the usefulness of the emotion fear in order to produce a change in intention and behavior to a "*plant-based*" diet among young adults in the United Kingdom, evoking environmental reasons.

Name company or University: De Haagse Hogeschool

External supervisor: Patricia Bulsing

UU Supervisor: Monique Smeets

Name Student: Nonea Alexa-Florina

Internship or Thesis: Internship

Title project: MOTIVATION AND REWARD IN A SOCIAL MEDIA ENVIRONMENT

Craicter is a reward based social media platform, a business at its beginning, which aims to combat the negative effects of social media through rewarding participants for their productive behavior. Users expectations were assured with a round of questionnaire was carried out. Firstly, to identify their motivating factors in relation to the use of social media. Secondly, to understand what kind of rewards will be more appealing and will motivate them more to interact with the app and get involved in social productive behavior.

The study concluded the possible ineffectiveness of promoting social activities as motivating and rewarding. With over 50% of users reluctant to use the platform's usefulness and efficiency, this study may represent an important fulcrum for improving Craicter's strategy and services.

Name company: Craciter

External supervisor: Tom Woods

UU Supervisor: Christoph Strauch

ABSTRACT

MOTIVATION AND REWARD IN A SOCIAL MEDIA ENVIRONMENT

By Nonea Alexa-Florina

Craicter is a reward based social media platform, a business at its beginning, which aims to combat the negative effects of social media through rewarding participants for their productive behavior. To assure a better understanding of user's behavior, a round of questionnaire was

carried out. Firstly, to identify their motivating factors in relation to the use of social media. Secondly, to understand what kind of rewards will be more appealing and will motivate them more to interact with the app and get involved in social productive behavior.

Through this survey, extrinsic motivation and real-life activities related rewards were correlated with a higher frequency of social media usage existed. At the same time, the high level of social media usage was positively correlated with the predisposition to get involved in rewarding or challenging activities.

Keywords: motivation, rewards, social media.

Name Student: Barry Kleijer

Internship or Thesis: Thesis

Title project: The Influence of Sensory Modality and the Quality of Sensory Information on Haptic Size Estimation

Haptic research is relevant for numerous applications where touch and kinesthetics are involved, such as telesurgery, operation of vehicles and virtual reality. The purpose of this study was to investigate the influence of sensory modality and the quality of sensory information on haptic size estimation. As stimuli, standardised cubes in the range of 3,8 to 6,2mm were used. Participants had to conduct a size estimation task in a high and low quality condition in the visual and haptic modality. The visual modality conditions were central or peripheral vision and for haptic condition, index/thumb or index/pink were used. Controlling of the visual condition was supported by webcam eye-tracking. Responses were measured in haptic modality on a Wacom tablet. Due to Covid measures, the study was conducted from a home-lab environment.

Name company or University: Utrecht University

UU Supervisor: Dr. Krista Overvliet

Abstract

Name Student: Zeinab Lalaoui El Mouttalibi

Internship or Thesis: Thesis

Title project: The subitization of multiple items and the extension of the subitization range across multiple fingers.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Krista Overvliet

In 1871 it was discovered that the human brain could enumerate up to 3 items at once. This type of enumeration is called subitizing. Previously it was thought that subitizing only occurred in vision, but studies in tactile enumeration have shown that subitizing also occurs through touch. This study will investigate whether subitizing still occurs across fingers when each finger is presented with multiple items and whether this increases the subitizing range. To answer the question 4 experiments were conducted. In the first experiment participants were not aware of the amount of items presented nor were they aware of which fingers were going to be stimulated. During the second experiment they were aware of the fingers that were going to be stimulated but not of how many items were going to be presented. Contrary to the second experiment they were aware of the amount of items but not which fingers were going to be stimulated during the third experiment. Lastly they were neither aware of the number of items presented nor which fingers were going to be stimulated.

Name Student: Lauren Travers

Internship or Thesis: Internship

Title project: “How to realise a medical device – an insight into my internship at ONWARD”

Our increasing daily interactions with technology is causing as much frustration as it is making our lives easier. This is mostly due to designers ignoring their user’s needs. When designing a medical device there are many different user groups with separate needs, as well as many standards and regulations to follow. Therefore, it is important to have a plan that incorporates human factors engineering processes and provides traceability. Here, I will introduce the V-Cycle, a comprehensive plan used to realise a medical device, and show how we can use this plan to design a medical device that is safe and reliable, as well as pleasurable to use for its intended users.

Name company or University: ONWARD Medical

(If applicable) External supervisor: Francesca Stradolini

UU Supervisor: Krista Overvliet

Name Student: Ayona Sengupta

Internship or Thesis: Internship

Title Project: Let's Take This "Online": A Comparison of Online and Offline Methods of Psychological Testing

Abstract

The development of technology and its widespread incorporation into households has marked the beginning of many industries and practices shifting "online". Similarly, many behavioural researchers have turned to online tools and web-based solutions. In order to explore the implementation of online tools in more sensitive experiments of cognition, such as those involving reaction time, a formal comparison of accuracy and precision was drawn between the online software Gorilla and the offline software PsychoPy using a Go/No Go task. Information on the accuracy and precision of the measured values for response time and error rates were documented, and a comparison of the absolute values measured and the spread of data between the two methods was completed. Analysis of the data has shown that both the online and offline methods perform at similar levels of accuracy and precision for groups completing the Go/No Go task, suggesting that both Gorilla and PsychoPy are viable platforms for more sensitive cognitive experiments.

Name company or University: Utrecht University

UU Supervisor: Dr.Jeroen Benjamins

Name Student: Ype de Boer

Internship or Thesis: Thesis

Title project: Elevating visibility of visually impaired pedestrians to increase stopping of cars at crossings at night

Abstract (100 - 150 words):

Over 300.000 people in the Netherlands are visually impaired. Their number one issue is in navigation and travel. They are heavily reliant on other road users to perceive them and behave accordingly. At night it is more difficult to be perceived which gives the following question: Which visual elements best increase visibility and as such best increase safety for visually impaired pedestrians in crossing the street at night? The question gets answered via real world tests with car drivers at night and by asking visually impaired pedestrians how they feel walking at night and if they prefer other times of day for safety reasons.

Name company or University: UU, Bartiméus

(If applicable) External supervisor:

UU Supervisor: Albert Postma

Name Student: Stein Bellers

Internship or Thesis: Thesis

Title project: Eye tracking without screens: potential problems and how to deal with them

In my thesis I provide an overview of the strengths and weaknesses of a screenless head boxed eye-tracking setup, the effectiveness of different calibration methods and insight into what to expect with regards to a phenomenon called the parallax error. This study is aimed at people that are looking into different methods of eye-tracking for their studies and want to know of the advantages and disadvantages of screenless head boxed eye-tracking and some tips to optimize the accuracy and precision of the data.

Name company or University: Utrecht university

UU Supervisor: Niilo Valtakari

Name Student: Fatih Celalettin Deniz

Internship or Thesis: Thesis

Title project: Reading Emotions: Automatic Recognition of Engagement Using Facial Expressions and Physiological Measures

Engagement is a state of being involved, occupied, or fully absorbed in something. Although this definition of engagement looks clear at the first glance, a closer look into the literature reveals the indistinctness of the concept. On the other hand, there is a consensus on the benefits of engagement regardless of its definition of it. Despite its importance, there are many methods for measuring engagement. This study utilizes both subjective and objective measures to investigate engagement by using facial expressions, skin conductance, heart rate, and gaze direction to discover how these measures correlate with engagement. This multimodal approach will reveal a broader picture for engagement.

Name company or University: Noldus Information Technology

External Supervisor: Tess den Uyl

UU Supervisor: Stella Donker

Name Student: Vitali Lambregts

Internship or Thesis: Thesis

Title project: Detecting cognitive fatigue with an impairment-assessment application such as DRUID using EEG

Abstract:

Even though common everyday tasks such as driving are in nature rather complex and require cognitive abilities to be carried out adequately, it is for most experienced drivers rather automatic behavior and requires little effort. However, vigilance or sustained attention is still important to be able to adequately respond to unexpected events on the road. Given that (sustained) attention and performance on complex tasks is negatively affected by mental fatigue. Driving while mentally fatigued is a serious problem, believed to be a large contributor traffic accidents. Therefore it is important to investigate options that can possible intervene with driving while mentally fatigued. In this experiment it is tested whether a quick 2-minute impairment-detection application (such as DRUID) can detect impairment resulting from mental fatigue after a cognitive demanding task (CTET). EEG data, as well as subjective measurements will be used to assess whether participants are subjected to mental fatigue.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Leon Kenemans

Name Student: Fie van Weering

Internship or Thesis: Thesis

Title project: Encouraging plant-based food choices among meat consumers: The effect of hedonic versus universalistic value communication

Previous studies suggest that taking into account people's motivational values can help facilitating the shift towards more plant-based diets. The values hedonism (*fulfilling individual needs*) and universalism (*protection and welfare of all people and nature*) appear to be involved in food choice. In an online study with a within-subject design, we examined whether giving arguments that target hedonic values (versus universalistic values) would lead to increased plant-based food attractiveness and plant-based food choice among meat consumers. The results indicate that promotion is a helpful way to increase plant-based food choices among meat consumers and that targeting hedonic values seems to be most effective. In addition, it is suggested that there might be a spill-over effect in which promotion increases the evaluation and food choice for plant-based items in general.

Name company or University: Behavioural Science Institute, Radboud University

(If applicable) External supervisor: Thijs Verwijmeren

UU Supervisor: Jeroen Benjamins